

Experiment 7

Name: S.Kathir

Reg. no.: 3122235002059

1. Aim

To develop a Student Management System using Ruby on Rails that allows users to:

- Add new students
- View student records
- Update student details
- Delete student records

The system should implement full CRUD operations with proper validation and error handling.

Required Web Tools and Methodology

Tools Used

- Ruby (3.2.2)
- Ruby on Rails (7.x)
- SQLite3 Database
- VS Code
- Web Browser (Chrome)

Methodology

- Followed MVC Architecture (Model-View-Controller)
- Used ActiveRecord for database interaction
- Used RESTful routing
- Implemented CRUD operations
- Added validations in model
- Styled UI using CSS

Implementation Procedure

1. Install Ruby and Rails.
2. Create a new Rails application:
`rails new StudentManagementSystem`

```
cd StudentManagementSystem
```

3. Generate Student model:

```
rails generate model Student name:string email:string age:integer
```

4. Run migration:

```
rails db:migrate
```

5. Generate controller:

```
rails generate controller Students
```

6. Define routes using:

```
resources :students
```

7. Implement CRUD actions in controller.

8. Create form using form_with.

9. Add validations in model.

10. Add CSS styling.

11. Run server:

```
rails s
```

Code:

Model (student.rb)

```
class Student < ApplicationRecord
  validates :name, presence: true
  validates :email, presence: true, uniqueness: true
  validates :age, numericality: { only_integer: true }, allow_nil: true
end
```

Controller (students_controller.rb)

```
class StudentsController < ApplicationController
  before_action :set_student, only: [ :edit, :update, :destroy, :show ]

  # GET /students
  def index
    @students = Student.all
  end
end
```

```
end
```

```
# GET /students/new
```

```
def new
```

```
  @student = Student.new
```

```
end
```

```
# POST /students
```

```
def create
```

```
  @student = Student.new(student_params)
```

```
  if @student.save
```

```
    redirect_to students_path, notice: "Student created successfully."
```

```
  else
```

```
    render :new, status: :unprocessable_entity
```

```
  end
```

```
end
```

```
# GET /students/:id/edit
```

```
def edit
```

```
end
```

```
# PATCH/PUT /students/:id
```

```
def update
```

```
  if @student.update(student_params)
```

```
    redirect_to students_path, notice: "Student updated successfully."
```

```
  else
```

```
    render :edit, status: :unprocessable_entity
```

```
  end
```

```
end
```

```
# DELETE /students/:id

def destroy

  @student.destroy

  redirect_to students_path, notice: "Student deleted successfully."
end


private


def set_student

  @student = Student.find(params[:id])
end


def student_params

  params.require(:student).permit(:name, :email, :age)
end


def show

end

end
```

Routes (routes.rb)

```
Rails.application.routes.draw do

  resources :students

  root "students#index"

end
```

Views

index.html.erb

```
<h1>Students</h1>
```

```
<div class="container">
```

```

<% flash.each do |type, message| %>

  <div class="flash-message <%= type %>">

    <%= message %>

  </div>

<% end %>

<%= link_to "Add New Student", new_student_path %>

<table border="1">

<tr>

  <th>Name</th>

  <th>Email</th>

  <th>Age</th>

  <th>Actions</th>

</tr>

<% @students.each do |student| %>

  <tr>

    <td><%= student.name %></td>

    <td><%= student.email %></td>

    <td><%= student.age %></td>

    <td>

      <%= link_to "Edit", edit_student_path(student) %> |

      <%= link_to "Delete", student_path(student), data: { turbo_method: :delete,
turbo_confirm: "Are you sure?" } %>

    </td>

  </tr>

<% end %>

</table>

</div>

```

_form.html.erb

```
<%= form_with(model: @student, local: true) do |form| %>
```

```
<% if @student.errors.any? %>
```

```
<div class="error-box">
```

```
<h3><%= pluralize(@student.errors.count, "error") %> prohibited this student from being  
saved:</h3>
```

```
<ul>
```

```
<% @student.errors.full_messages.each do |message| %>
```

```
<li><%= message %></li>
```

```
<% end %>
```

```
</ul>
```

```
</div>
```

```
<% end %>
```

```
<div>
```

```
<%= form.label :name %><br>
```

```
<%= form.text_field :name %>
```

```
</div>
```

```
<div>
```

```
<%= form.label :email %><br>
```

```
<%= form.email_field :email %>
```

```
</div>
```

```
<div>
```

```
<%= form.label :age %><br>
```

```
<%= form.number_field :age %>
```

```
</div>
```

```
<div>
```

```
<%= form.submit %>
```

```
</div>
```

```
<% end %>
```

new.html.erb

```
<h1>New Student</h1>
```

```
<div class="container">
```

```
  <%= render 'form' %>
```

```
  <%= link_to "Back", students_path %>
```

```
</div>
```

edit.html.erb

```
<h1>Edit Student</h1>
```

```
<div class="container">
```

```
  <%= render 'form' %>
```

```
  <%= link_to "Back", students_path %>
```

```
</div>
```

show.html.erb

```
<h1>Student Details</h1>
```

```
<p><strong>Name:</strong> <%= @student.name %></p>
```

```
<p><strong>Email:</strong> <%= @student.email %></p>
```

```
<p><strong>Age:</strong> <%= @student.age %></p>
```

```
<%= link_to "Back", students_path %>
```

application.css

```
/* GENERAL PAGE STYLING */
```

```
body{  
    font-family: Arial, Helvetica, sans-serif;  
    background-color: #f4f6f9;  
    margin: 0;  
    padding: 0;  
}
```

```
h1 {  
    text-align: center;  
    margin-top: 40px;  
    color: #333;  
}
```

```
/* CONTAINER */
```

```
.container {  
    width: 40%;  
    margin: 40px auto;  
    background: white;  
    padding: 30px 30px;  
    border-radius: 10px;  
    box-shadow: 0 4px 12px rgba(0,0,0,0.1);  
}
```

```
/* FORM STYLING */
```

```
form div {  
    margin-bottom: 15px;  
}
```



```
form {  
  margin: 0;  
}
```

```
label {  
  font-weight: bold;  
}
```

```
input[type="text"],  
input[type="email"],  
input[type="number"] {  
  width: 100%;  
  padding: 8px;  
  margin-top: 5px;  
  border: 1px solid #ccc;  
  border-radius: 5px;  
  box-sizing: border-box;  
}
```

```
input[type="submit"] {  
  background-color: #4CAF50;  
  color: white;  
  padding: 10px 15px;  
  border: none;  
  border-radius: 5px;  
  cursor: pointer;  
}
```

```
input[type="submit"]:hover {  
  background-color: #45a049;  
}
```

```
/* TABLE STYLING */
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
    margin-top: 20px;  
}
```

```
table th {  
    background-color: #4CAF50;  
    color: white;  
    padding: 10px;  
}
```

```
table td {  
    padding: 10px;  
    text-align: center;  
    border-bottom: 1px solid #ddd;  
}
```

```
table tr:hover {  
    background-color: #f1f1f1;  
}
```

```
/* LINKS */
```

```
a {  
    text-decoration: none;  
    color: #007bff;  
    font-weight: bold;  
}
```

```
a:hover {  
    text-decoration: underline;  
}
```

```
/* ERROR BOX */
```

```
.error-box {  
    background-color: #ffe6e6;  
    border: 1px solid red;  
    padding: 10px;  
    border-radius: 5px;  
    margin-bottom: 15px;  
    color: red;  
}
```

```
/* BUTTON LINKS */
```

```
.button-link {  
    background-color: #007bff;  
    color: white;  
    padding: 6px 10px;  
    border-radius: 4px;  
}
```

```
.button-link:hover {  
    background-color: #0056b3;  
}
```

```
.flash-message {  
    width: 40%;  
    margin: 20px auto; /* This centers it */  
    padding: 12px 15px;  
    border-radius: 8px;
```

```
text-align: center;

font-weight: 600;

}


/* Success message */

.flash-message.notice {

background-color: #d4edda;

color: #155724;

border: 1px solid #c3e6cb;

}


/* Alert / error message */

.flash-message.alert {

background-color: #f8d7da;

color: #721c24;

border: 1px solid #f5c6cb;

}
```

Output:

Add New Student Page

Submitting with blank fields

A screenshot of a web browser displaying a 'New Student' form. The browser's address bar shows '127.0.0.1:3000/students/new'. The form is titled 'New Student' and contains three input fields: 'Name', 'Email', and 'Age'. A red error box at the top of the form states: '2 errors prohibited this student from being saved:'. Below this, a list of errors is shown: 'Name can't be blank' and 'Email can't be blank'. At the bottom of the form, there is a green 'Create Student' button and a blue 'Back' link.

127.0.0.1:3000/students/new

New Student

2 errors prohibited this student from being saved:

- Name can't be blank
- Email can't be blank

Name

Email

Age

Create Student

[Back](#)

Validation

A screenshot of a web browser displaying a 'New Student' form. The browser's address bar shows '127.0.0.1:3000/students/new'. The form is titled 'New Student' and contains three input fields: 'Name', 'Email', and 'Age'. The 'Name' field contains the text 'Kathir'. The 'Email' field contains the text 'kathir2310902'. The 'Age' field contains the text '20'. A validation error message is displayed below the 'Email' field: 'Please include an '@' in the email address. 'kathir2310902' is missing an '@'. At the bottom of the form, there is a green 'Create Student' button and a blue 'Back' link.

127.0.0.1:3000/students/new

New Student

Name

Email

Age

Please include an '@' in the email address. 'kathir2310902' is missing an '@'.

Create Student

[Back](#)

127.0.0.1:3000/students/new

All Bookmarks

New Student

Name

Email

Age

Create Student

[Back](#)

View Students Page

127.0.0.1:3000/students

All Bookmarks

Students

Student created successfully.

[Add New Student](#)

Name	Email	Age	Actions
Kathir	kathir2310902@ssn.edu.in	20	Edit Delete

Edit Student Page

127.0.0.1:3000/students/11/edit

All Bookmarks

Edit Student

Name

S Kathir

Email

kathir2310902@ssn.edu.in

Age

20

Update Student

Back

After editing student details:

127.0.0.1:3000/students

All Bookmarks

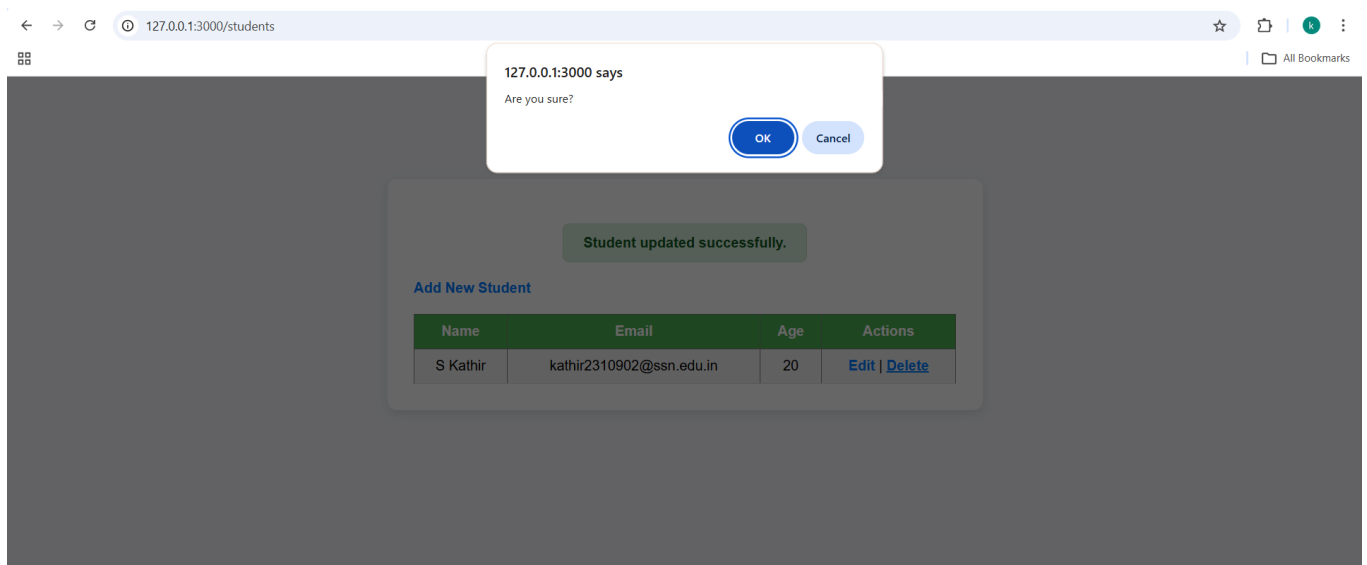
Students

Student updated successfully.

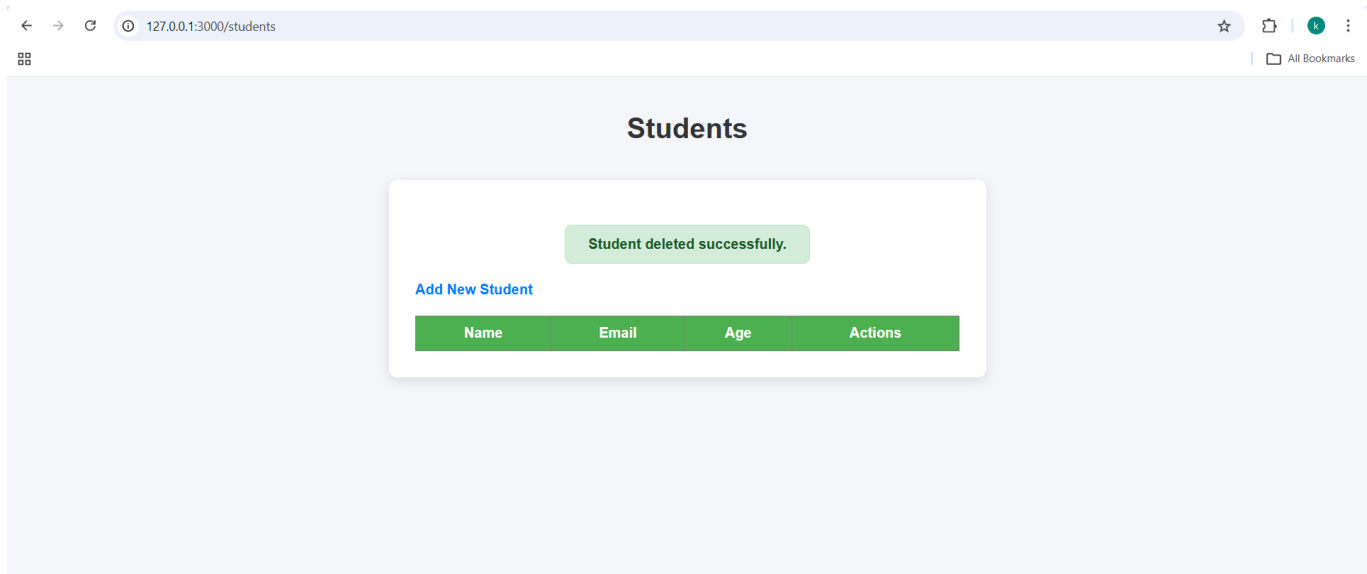
[Add New Student](#)

Name	Email	Age	Actions
S Kathir	kathir2310902@ssn.edu.in	20	Edit Delete

Deleting a student:



After deletion:



Conclusion

The Student Management System was successfully developed using Ruby on Rails following MVC architecture. The application performs full CRUD operations with proper validation and error handling. The project demonstrates effective use of RESTful routing, ActiveRecord, and form handling in Rails.

2. Aim

To develop a Library Management System using Ruby on Rails that allows users to:

- Add new books
- View list of books
- View details of a single book
- Update book information
- Delete book records

The system should implement full CRUD operations with proper validation and error handling.

Required Web Tools and Methodology

Tools Used

- Ruby (3.2.x)
- Ruby on Rails (7.x)
- SQLite3 Database
- VS Code
- Web Browser (Chrome/Edge)

Methodology

The application follows the MVC (Model-View-Controller) architecture:

- Model: Handles data and validations
- View: Handles user interface
- Controller: Handles business logic and user requests

The application uses:

- RESTful routing
- ActiveRecord for database operations
- form_with for form handling
- Model validations for error handling

Implementation Procedure

Step 1: Create Rails Application

```
rails new LibraryManagementSystem
```

```
cd LibraryManagementSystem
```

Step 2: Generate Book Model

`rails generate model Book title:string author:string published_year:integer`

Step 3: Run Migration

`rails db:migrate`

This creates the books table in the database.

Step 4: Generate Controller

`rails generate controller Books`

Step 5: Configure Routes

Open `config/routes.rb`:

`resources :books`

`root "books#index"`

Step 6: Add Validations in Model

Add presence validation and year validation.

Step 7: Implement CRUD Methods in Controller

Implement:

- `index`
- `show`
- `new`
- `create`
- `edit`
- `update`
- `destroy`

Step 8: Create Views

Create:

- index page
- show page
- new page
- edit page
- reusable form partial

Step 9: Add CSS Styling

Add styling in application.css.

Step 10: Run Server

rails s

Code

Model (app/models/book.rb)

```
class Book < ApplicationRecord
  validates :title, presence: true
  validates :author, presence: true
  validates :published_year,
    presence: true,
    numericality: { only_integer: true },
    comparison: { less_than_or_equal_to: ->(_) { Date.current.year }}
end
```

Controller (app/controllers/books_controller.rb)

```
class BooksController < ApplicationController
  before_action :set_book, only: [:show, :edit, :update, :destroy]

  def index
    @books = Book.all
  end
```

```
def show
```

```
end
```

```
def new
```

```
  @book = Book.new
```

```
end
```

```
def create
```

```
  @book = Book.new(book_params)
```

```
  if @book.save
```

```
    redirect_to books_path, notice: "Book created successfully."
```

```
  else
```

```
    render :new, status: :unprocessable_entity
```

```
  end
```

```
end
```

```
def edit
```

```
end
```

```
def update
```

```
  if @book.update(book_params)
```

```
    redirect_to books_path, notice: "Book updated successfully."
```

```
  else
```

```
    render :edit, status: :unprocessable_entity
```

```
  end
```

```
end
```

```
def destroy
```

```
  @book.destroy
```

```
  redirect_to books_path, notice: "Book deleted successfully."
```

```
end
```

```
private
```

```
def set_book
```

```
  @book = Book.find(params[:id])
```

```
end
```

```
def book_params
```

```
  params.require(:book).permit(:title, :author, :published_year)
```

```
end
```

```
end
```

Form (_form.html.erb)

```
<%= form_with(model: @book, local: true) do |form| %>
```

```
<% if @book.errors.any? %>
```

```
  <div class="error-box">
```

```
    <ul>
```

```
      <% @book.errors.full_messages.each do |message| %>
```

```
        <li><%= message %></li>
```

```
      <% end %>
```

```
    </ul>
```

```
  </div>
```

```
<% end %>
```

```
<div>
```

```
  <%= form.label :title %>
```

```
  <%= form.text_field :title %>
```

```
</div>
```

```
<div>
```

```
<%= form.label :author %>

<%= form.text_field :author %>

</div>
```

```
<div>

  <%= form.label :published_year %>

  <%= form.number_field :published_year %>

</div>
```

```
<div>

  <%= form.submit %>

</div>
```

```
<% end %>
```

Index Page (index.html.erb)

```
<h1>Books</h1>
```

```
<div class="container">

  <% flash.each do |type, message| %>

    <div class="flash-message <%= type %>">

      <%= message %>

    </div>

  <% end %>
```

```
<%= link_to "Add New Book", new_book_path %>
```

```
<table border="1">

  <tr>

    <th>Title</th>

    <th>Author</th>
```

<th>Year</th>

<th>Actions</th>

</tr>

<% @books.each do |book| %>

<tr>

<td><%= link_to book.title, book_path(book) %></td>

<td><%= book.author %></td>

<td><%= book.published_year %></td>

<td>

<%= link_to "Edit", edit_book_path(book) %> |

<%= link_to "Delete", book_path(book),

data: { turbo_method: :delete, turbo_confirm: "Are you sure?" } %>

</td>

</tr>

<% end %>

</table>

</div>

Show Page (show.html.erb)

<h1>Book Details</h1>

<div class="container">

<p>Title: <%= @book.title %></p>

<p>Author: <%= @book.author %></p>

<p>Published Year: <%= @book.published_year %></p>

<%= link_to "Edit", edit_book_path(@book) %> |

<%= link_to "Back", books_path %>

</div>

Views

_form.html.erb

```
<%= form_with(model: @book, local: true) do |form| %>
```

```
  <% if @book.errors.any? %>
```

```
    <div class="error-box">
```

```
      <h3><%= pluralize(@book.errors.count, "error") %> prohibited this book from being
saved:</h3>
```

```
      <ul>
```

```
        <% @book.errors.full_messages.each do |message| %>
```

```
          <li><%= message %></li>
```

```
        <% end %>
```

```
      </ul>
```

```
    </div>
```

```
  <% end %>
```

```
<div>
```

```
  <%= form.label :title %>
```

```
  <%= form.text_field :title %>
```

```
</div>
```

```
<div>
```

```
  <%= form.label :author %>
```

```
  <%= form.text_field :author %>
```

```
</div>
```

```
<div>
```

```
  <%= form.label :published_year %>
```

```
  <%= form.number_field :published_year %>
```

```
</div>
```



```
<div>
```

```
  <%= form.submit %>
```

```
</div>
```

```
<% end %>
```

edit.html.erb

```
<h1>Edit Book</h1>
```

```
<div class="container">
```

```
  <%= render 'form' %>
```

```
  <%= link_to "Back", books_path %>
```

```
</div>
```

index.html.erb

```
<h1>Books</h1>
```

```
<div class="container">
```

```
  <% flash.each do |type, message| %>
```

```
    <div class="flash-message <%= type %>">
```

```
      <%= message %>
```

```
    </div>
```

```
  <% end %>
```

```
<%= link_to "Add New Book", new_book_path %>
```

```
<table border="1">
```

```
  <tr>
```

```
    <th>Title</th>
```

```
    <th>Author</th>
```

```
    <th>Year</th>
```

```
<th>Actions</th>
```

```
</tr>
```

```
<% @books.each do |book| %>
```

```
<tr>
```

```
<td><%= link_to book.title, book_path(book) %></td>
```

```
<td><%= book.author %></td>
```

```
<td><%= book.published_year %></td>
```

```
<td>
```

```
<%= link_to "Edit", edit_book_path(book) %> |
```

```
<%= link_to "Delete", book_path(book),
```

```
  data: { turbo_method: :delete, turbo_confirm: "Are you sure?" } %>
```

```
</td>
```

```
</tr>
```

```
<% end %>
```

```
</table>
```

```
</div>
```

new.html.erb

```
<h1>New Book</h1>
```

```
<div class="container">
```

```
<%= render 'form' %>
```

```
<%= link_to "Back", books_path %>
```

```
</div>
```

show.html.erb

```
<h1>Book Details</h1>
```

```
<div class="container">
```

```
<p><strong>Title:</strong> <%= @book.title %></p>
```

```
<p><strong>Author:</strong> <%= @book.author %></p>

<p><strong>Published Year:</strong> <%= @book.published_year %></p>

<%= link_to "Edit", edit_book_path(@book) %> |

<%= link_to "Back", books_path %>

</div>
```

application.css

```
/* BODY */

body {

  font-family: Arial, Helvetica, sans-serif;

  background-color: #f4f6f9;

  margin: 0;

  padding: 0;

}


/* HEADINGS */

h1 {

  text-align: center;

  margin-top: 40px;

  color: #2c3e50;

}


/* CONTAINER (CARD STYLE) */

.container {

  width: 45%;

  margin: 40px auto;

  background: #ffffff;

  padding: 30px 40px;

  border-radius: 12px;

  box-shadow: 0 8px 20px rgba(0, 0, 0, 0.08);

}
```

```
}
```

```
/* FORM STYLING */
```

```
form {
```

```
    margin: 0;
```

```
}
```

```
form div {
```

```
    margin-bottom: 20px;
```

```
}
```

```
label {
```

```
    font-weight: 600;
```

```
    color: #333;
```

```
}
```

```
/* INPUT FIELDS */
```

```
input[type="text"],
```

```
input[type="number"] {
```

```
    width: 100%;
```

```
    padding: 10px;
```

```
    margin-top: 6px;
```

```
    border: 1px solid #ccc;
```

```
    border-radius: 6px;
```

```
    box-sizing: border-box;
```

```
    transition: 0.3s;
```

```
    font-size: 14px;
```

```
}
```

```
/* Input focus effect */
```

```
input[type="text"]:focus,
```

```
input[type="number"]:focus {  
    border-color: #4CAF50;  
    outline: none;  
    box-shadow: 0 0 5px rgba(76, 175, 80, 0.4);  
}
```

```
/* SUBMIT BUTTON */
```

```
input[type="submit"] {  
    background-color: #305ba0;  
    color: white;  
    padding: 10px 18px;  
    border: none;  
    border-radius: 6px;  
    cursor: pointer;  
    font-size: 14px;  
    transition: 0.3s;  
}
```

```
input[type="submit"]:hover {  
    background-color: #284c86;  
    transform: scale(1.03);  
}
```

```
/* TABLE STYLING */
```

```
table {  
    width: 100%;  
    border-collapse: collapse;  
    margin-top: 20px;  
}
```

```
table th {
```

```
background-color: #305ba0;
color: white;
padding: 10px;
text-align: center;
}
```

```
table td {
padding: 10px;
text-align: center;
border-bottom: 1px solid #ddd;
}
```

```
table tr:hover {
background-color: #f1f1f1;
}
```

```
/* LINKS */
```

```
a {
text-decoration: none;
font-weight: 600;
color: #007bff;
}
```

```
a:hover {
text-decoration: underline;
}
```

```
/* ERROR BOX */
```

```
.error-box {
background-color: #f8d7da;
color: #721c24;
```

```
padding: 12px;
border-radius: 6px;
margin-bottom: 20px;
border: 1px solid #f5c6cb;
}
```

```
/* FLASH MESSAGES */
```

```
.flash-message {
width: 45%;
margin: 20px auto;
padding: 12px;
border-radius: 8px;
text-align: center;
font-weight: 600;
/* box-shadow: 0 4px 10px rgba(0,0,0,0.08); */
}
```

```
/* Success message */
```

```
.flash-message.notice {
background-color: #d4edda;
color: #155724;
border: 1px solid #c3e6cb;
}
```

```
/* Alert message */
```

```
.flash-message.alert {
background-color: #f8d7da;
color: #721c24;
border: 1px solid #f5c6cb;
}
```

Output:

Add New Book Page

Validation

A browser window showing the 'New Book' form at the URL 127.0.0.1:3000/books/new. The form contains three input fields: 'Title' with the value 'The Interpreter of Maladies', 'Author' with the value 'Jhumpa Lahiri', and 'Published year' with the value '2027'. A red error message box at the top of the form states: '1 error prohibited this book from being saved: • Published year must be less than or equal to 2026'. Below the inputs are a blue 'Create Book' button and a blue 'Back' link.

New Book

1 error prohibited this book from being saved:

- Published year must be less than or equal to 2026

Title
The Interpreter of Maladies

Author
Jhumpa Lahiri

Published year
2027

Create Book

[Back](#)

A browser window showing the 'New Book' form at the URL 127.0.0.1:3000/books/new. The form contains three input fields: 'Title' with the value 'The Interpreter of Maladies', 'Author' with the value 'Jhumpa Lahiri', and 'Published year' with the value '2000'. Below the inputs are a blue 'Create Book' button and a blue 'Back' link.

New Book

Title
The Interpreter of Maladies

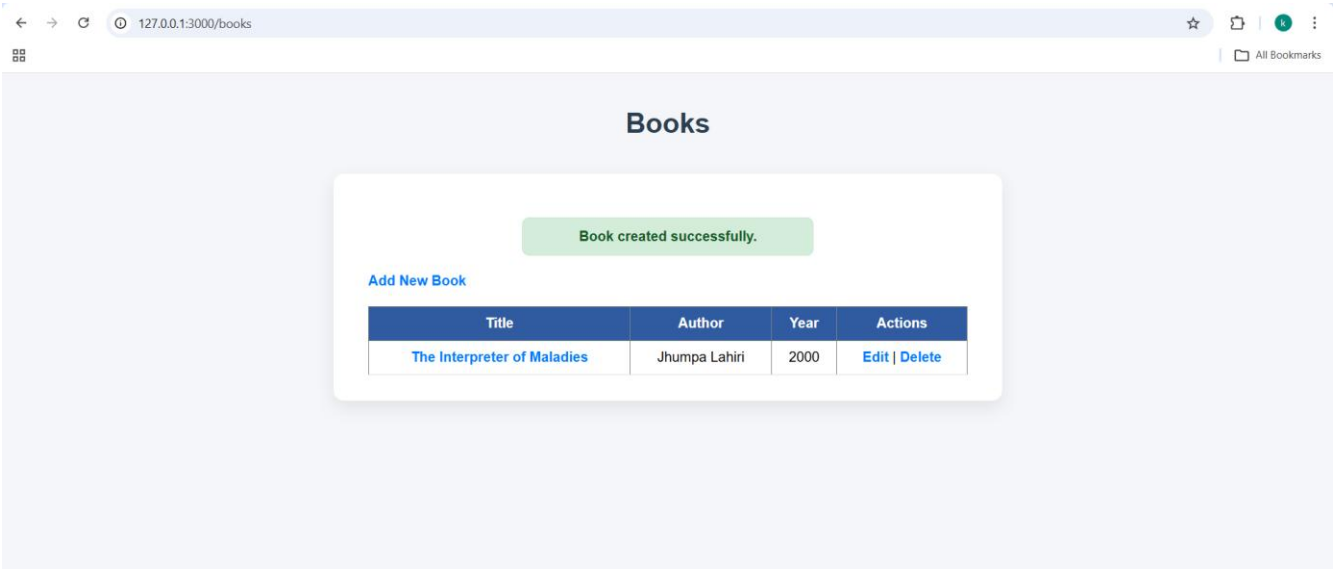
Author
Jhumpa Lahiri

Published year
2000

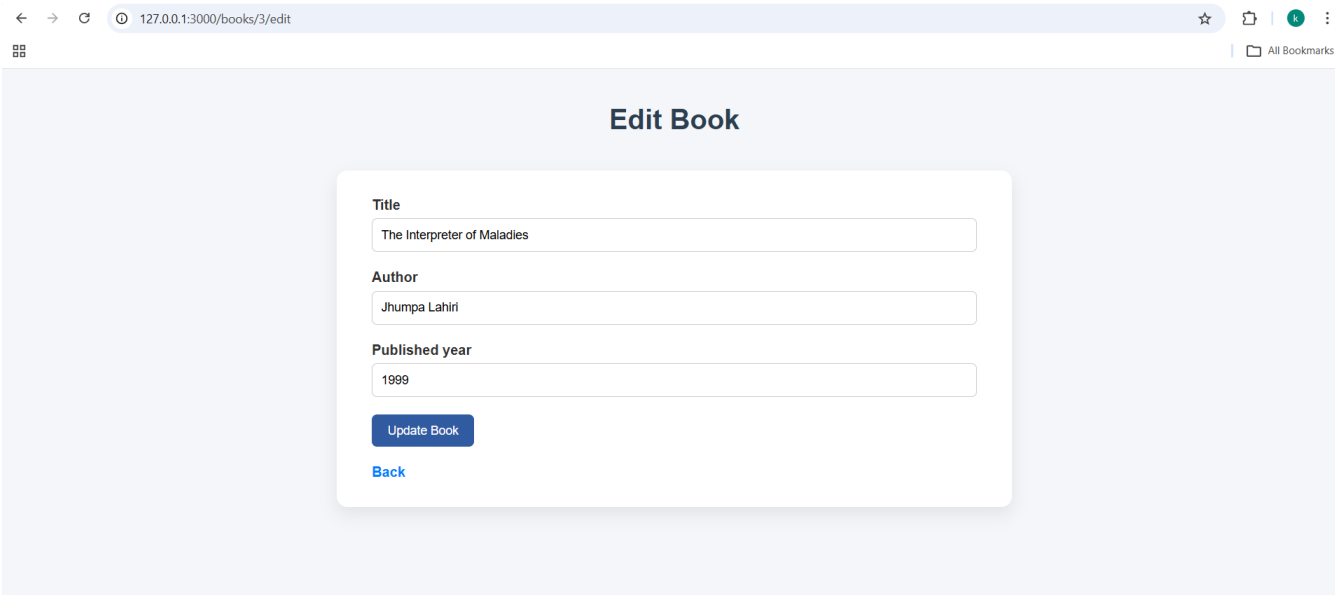
Create Book

[Back](#)

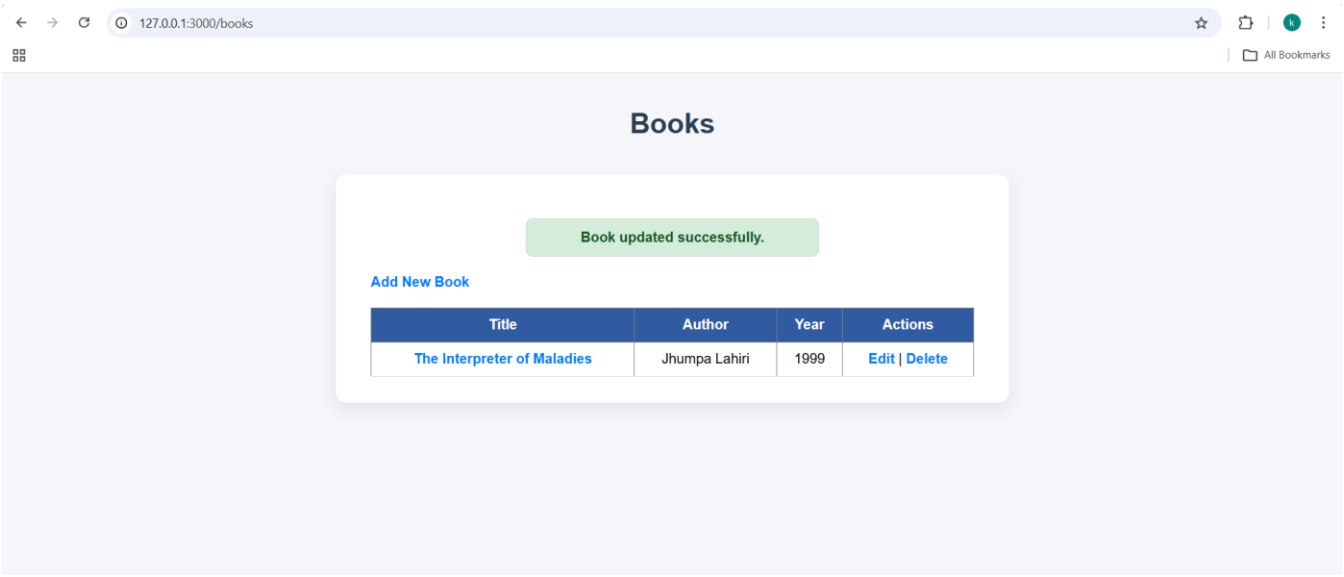
Index Page



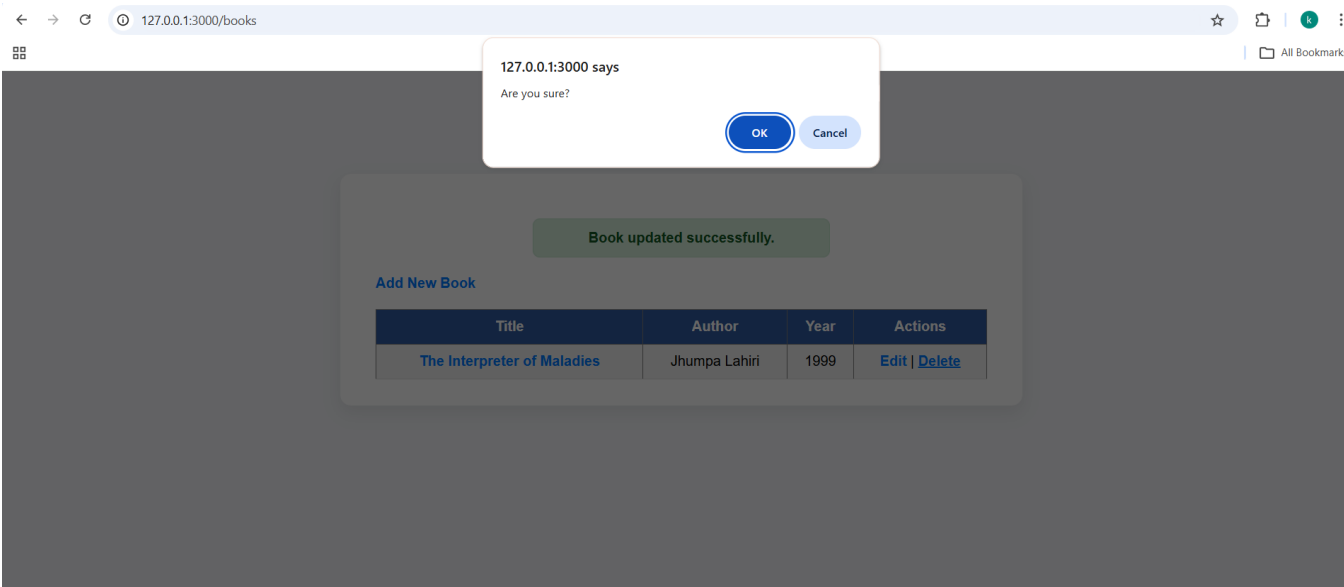
Edit Book Page

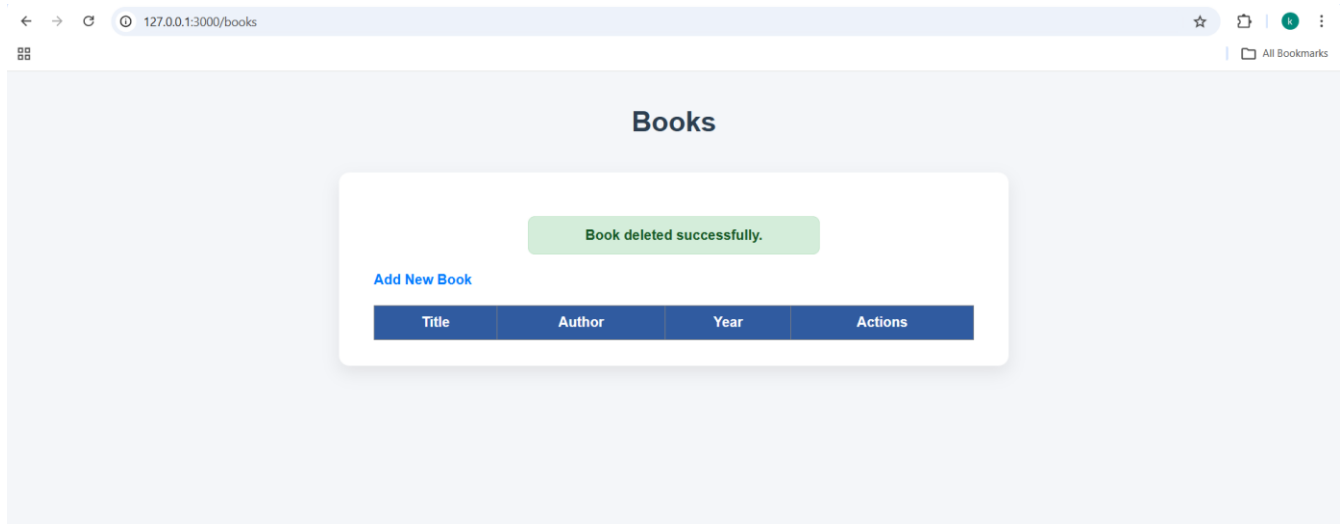


After editing book details

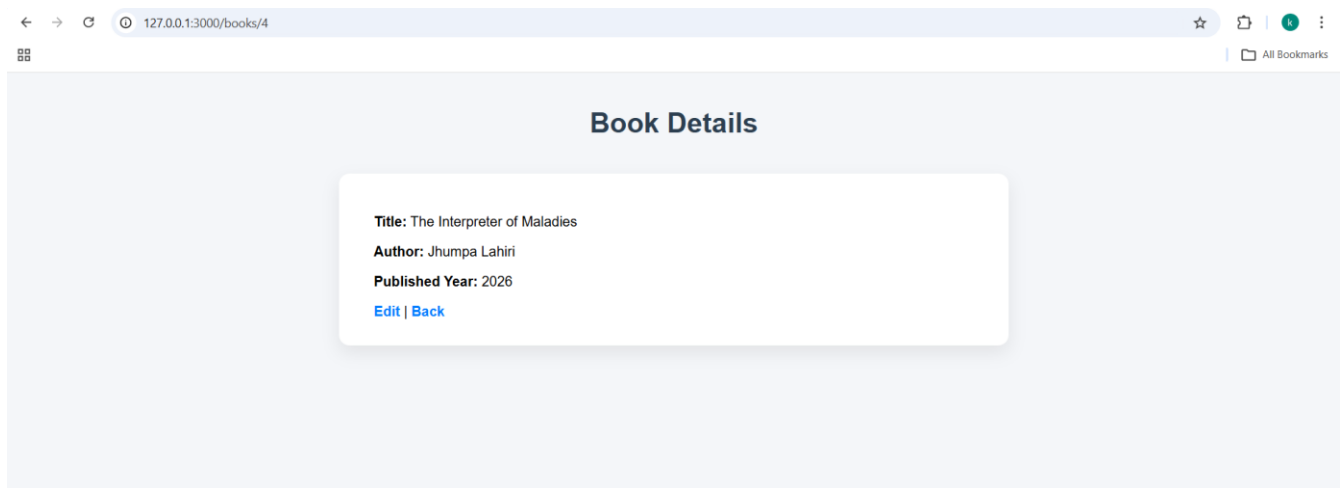


Deletion





Show Page



Conclusion

The Library Management System was successfully implemented using Ruby on Rails following MVC architecture. The application supports complete CRUD operations with proper validation and error handling. RESTful routing and ActiveRecord validations ensure efficient data management and integrity. The system demonstrates effective use of Rails framework features in developing a structured web application.